10. The Spark Gap

Read, model, and verify only — produce no sparks

Gas breakdown as historical physics, not a learner build

Lesson 10 · about 60 minutes · paper evidence only

Stop line: this is an interpretation lesson

Learner boundary: There is no learner apparatus in this chapter. Do not improvise a gap, charge a capacitor, use a piezoelectric igniter, open a lighter, approach an overhead line, or connect any historical device. Photographs, printed graphs, and the drawings here are the evidence. A museum barrier is part of the apparatus. Mains, exposed high voltage, charged power capacitors, ignition parts, and spark-producing apparatus remain outside this curriculum.

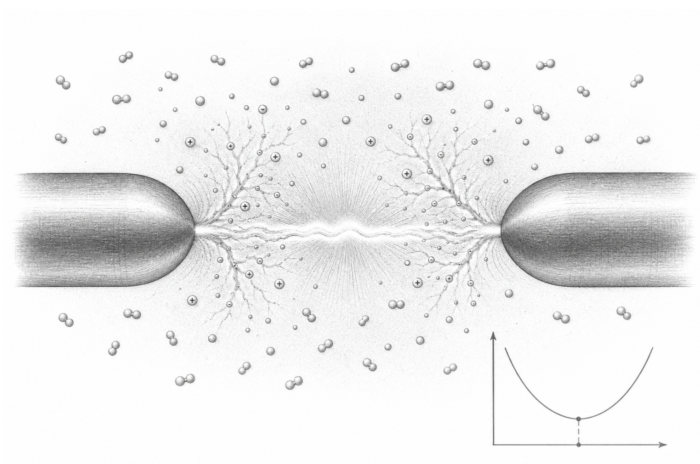
What you need

- This reading and a pencil
- The printed graph and historical-apparatus plates in this chapter
- No energized apparatus, loose electrodes, capacitors, or ignition parts

Predict first

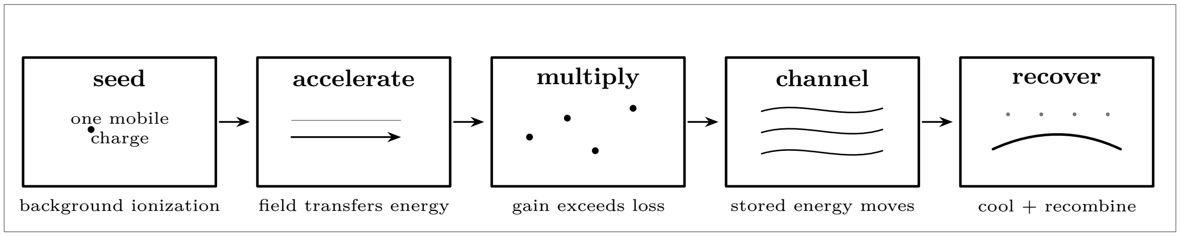
Air is often called an insulator. What observation would show that this is a useful approximation rather than an unconditional property?

Read the event without making one



Conceptual graphite cutaway. It identifies stages, not dimensions, parts, or a construction method.

Before breakdown, a few mobile charges already exist through background ionization. An electric field accelerates them. If a collision transfers enough energy to free another electron, the number of charge carriers can grow. Losses to walls, attachment, and recombination compete with that growth. A self- sustaining avalanche creates a conducting channel; stored energy can then move through it rapidly. The channel heats, glows, expands, and eventually recovers as charge density and temperature fall.

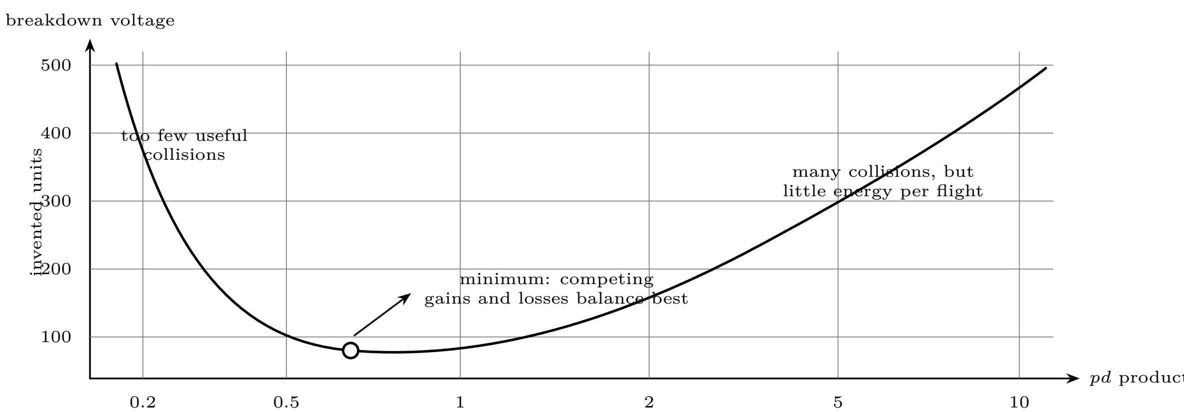


A causal sequence, not a recipe. Time and distance are not to scale.

Necessary	An electric field, seed charge carriers, and collision conditions that permit net multiplication.
Not sufficient	A visible gap by itself. Gas, pressure, electrode shape, contamination, polarity, and history also matter.
What light proves	Excited atoms and molecules released photons. It does not by itself reveal voltage, current, or delivered energy.
What sound proves	The gas changed pressure rapidly. Loudness is not a safe proxy for electrical energy.

Read a breakdown curve

For a fixed gas and electrode system, a Paschen-style graph relates breakdown voltage to the product of pressure p and separation d. The axes below are deliberately dimensionless and the values are invented for graph-reading practice. They are not design data.



Practice curve only. Never use it to choose a voltage, gap, gas, or pressure.

Worked reading 1: same distance, different pressure

Two sealed historical tubes have the same separation. Tube A is plotted at $pd = 0.5$; tube B at $pd = 5$. Reading vertically from the practice curve gives about 130 invented voltage units for A and about 320 for B. The separation did not change, yet the breakdown condition did. Therefore "gap alone sets the voltage" is falsified by this pair.

Worked reading 2: two products, one approximate voltage

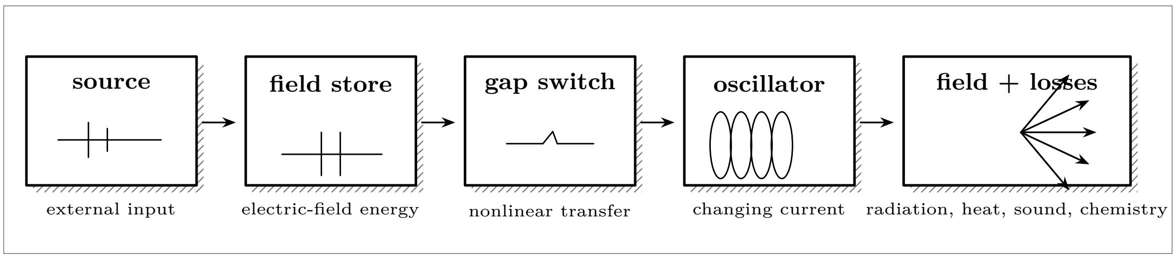
A horizontal line at 200 invented units crosses the curve on both sides of the minimum. One reading has relatively few effective collisions; the other has frequent collisions but shorter energy-gaining flights. A single observed breakdown voltage therefore need not identify a unique pd product.

Graph-reading questions

1. At the marked minimum, would moving either left or right lower the breakdown value? *No. Both directions raise it on this model.*
2. If pressure doubles and distance halves, what happens to pd? *It stays constant; this curve predicts the same point, within its scope.*
3. Does the curve predict all electrode shapes and gas mixtures? *No. Those conditions define which curve is relevant.*
4. A datum lies far from the curve. Is "bad measurement" the only explanation? *No: wrong gas, pressure error, surface condition, geometry, or an inadequate model are competing explanations.*

Trace energy through historical apparatus

The plate below abstracts a museum spark transmitter. It omits values, connections, dimensions, adjustment methods, and operating procedure. Its purpose is to keep energy accounting separate from spectacle.



Energy-accounting plate. The boxes are functions, not an assembly diagram.

The gap supplies no energy. It changes state and permits previously stored field energy to move into a rapidly changing current. Some energy enters the intended electromagnetic oscillation; some becomes electrode heating, hot gas, light, sound, chemical products, and unintended radio emission. Historical utility does not imply present safety or spectrum compatibility.

Worked energy account

Suppose a museum placard says that 10 abstract energy units were stored before an event. A calorimetric reconstruction attributes 4 units to heating, 1 to light and sound, and 3 to the intended oscillation. The missing 2 units are not "destroyed." They set a measurement question: unmeasured radio emission, apparatus heating, residual stored energy, or uncertainty. A defensible claim is *accounted energy is 8 units*; it is not *efficiency is exactly 30%* until the system boundary and uncertainties are stated.

Claims that evidence can defeat

Claim	Discriminating observation	Conclusion
Gap alone determines breakdown	Same separation and gas, controlled pressure changed; breakdown value changed	Gap-only model is false.
The spark creates energy	Stored energy is reduced while heat, light, sound, and field energy appear	The gap redirects stored energy.
Brighter means more voltage	Brightness changes with current duration or camera exposure at comparable voltage	Brightness is not a voltmeter.
One curve fits every gas	A second gas produces a reproducibly shifted curve under matched geometry	Gas identity belongs in the model.
Every failed event proves insulation	Independent field and pressure checks show conditions were below the model's threshold	Only that tested condition is supported.

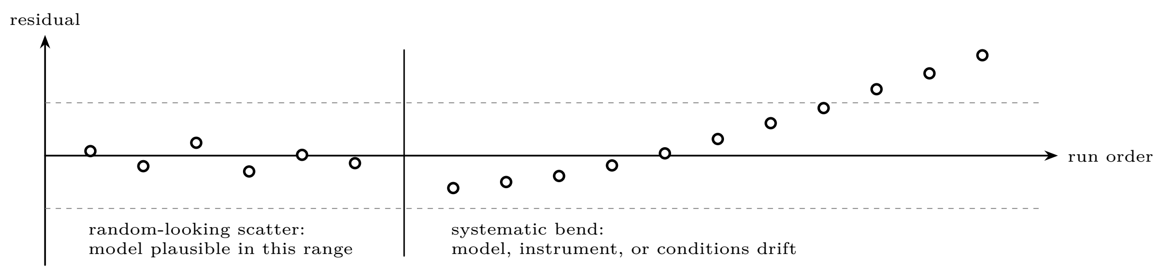
A SAFE FALSIFICATION BOUNDARY

Learners falsify claims by comparing supplied records, reading graphs, checking units, and identifying alternative explanations. They do not "test the threshold," shorten a gap, increase a supply, charge a store, or defeat a museum enclosure. A result that requires exposure to electrical danger is not a learner result.

Verify a reported result

Use this sequence on a paper dataset:

1. **Scope.** Record gas, pressure measure, separation measure, electrode description, polarity, temperature, and stated uncertainty.
2. **Axes.** Check whether the horizontal coordinate is p, d, or their product, and whether either axis is logarithmic.
3. **Replicates.** Look for spread, conditioning, and order effects; one event is not a curve.
4. **Competing accounts.** Ask what contamination, surface changes, pressure drift, or instrument limits would predict.
5. **Residual.** Plot observed minus modeled values. Random scatter and a systematic bend tell different stories.
6. **Claim.** State only what the measured range supports.



Residual patterns test the model more clearly than a smooth curve drawn through every point.

Worked verification question

A report gives ten breakdown values but omits pressure and electrode geometry. Can the Paschen-style model be verified? *No.* The values may be real, but the report lacks variables needed to locate the observations on the model and to decide whether the same curve applies. The correct next step is to request metadata, not to reproduce the event.

Change one thing — on paper

Choose one supplied curve point. Predict how its plotted position changes if pressure doubles while separation is halved. Then write:

1. the quantity held constant;
2. one assumption the graph requires;
3. one observation that would falsify your prediction; and
4. one reason the observation must come from a qualified laboratory or published dataset rather than a learner-built apparatus.

Evidence record

Prediction + reason ________________________________

Falsifier + boundary ________________________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University). Full citations and scope notes are recorded in the provider-neutral electricity research library.